

Title

Sentiment Analysis of Social Media Data

Objective

The objective of this project is to analyze sentiment patterns in social media posts to understand public opinion toward different brands using Natural Language Processing (NLP) and data visualization.

Dataset

Twitter Training Dataset

Columns:

- Tweet ID
 - Brand
 - Sentiment
 - Tweet
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Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- WordCloud

- Jupyter Notebook
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Data Preprocessing

- Loaded dataset
 - Removed missing values
 - Cleaned tweets
 - Removed URLs
 - Removed special characters
 - Converted text to lowercase
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Exploratory Data Analysis

Generated:

- Sentiment Distribution
 - Brand Distribution
 - Sentiment by Brand
 - Tweet Length Distribution
 - Word Clouds
 - Top Frequent Words
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Key Insights

- Positive and Negative sentiments dominate the dataset.
- Different brands receive varying public opinions.

- Word Clouds reveal frequently used terms in positive and negative tweets.
 - Tweet length varies significantly across sentiments.
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Conclusion

The project demonstrates how sentiment analysis can help organizations understand customer opinions and brand perception. Visualizations and text preprocessing provide meaningful insights from unstructured social media data.

Skills Learned

- Data Cleaning
- Natural Language Processing
- Text Preprocessing
- WordCloud
- Sentiment Analysis
- Exploratory Data Analysis
- Data Visualization